

Q. What is the difference between a microprocessor and a microcontroller?

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A. Microprocessors and microcontrollers differ in terms of their architecture as well as functionality.

A microprocessor is a programmable digital electronic component that incorporates the functions of a central processing unit (CPU) on a single semiconducting integrated circuit. It has the ability to execute a stored

general-purpose register (A, B, C, D, E, H and L), instruction register, interrupt control, serial input/output (I/O) control, program counter, stack pointer, flags (Z, S, P, CY and AC), clock, and data and address buses (8-, 16-, 32-bit and above).

The microprocessor by itself is incomplete. It has no memory, no input/output (I/O) port and no timer.

On the other hand, a microcontroller is a small (micro) single-chip computer designed to perform a specific function; the specific function

a microcontroller can also have a number of inbuilt peripherals such as analogue-to-digital converter, EEPROM, analogue comparators, capture-compare-PWM peripherals, serial synchronous and USART ports for serial communication, and parallel sync port.

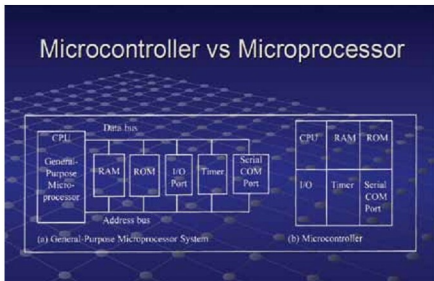
Microprocessors work with clock in GHz range, have a wide bus width and are expensive, whereas microcontrollers work with clock in the kHz-MHz range, have a shorter bus width and are inexpensive.

Microcontrollers add computing power to a device, enabling it to perform better at a low cost. Small packaging and low power requirements make these ideal for applications in which cost, power and space are crucial. Also, microcontrollers easily integrate with circuits, are programmable and reprogrammable, and have lots of inbuilt I/O capabilities.

In short, a microprocessor is a general-purpose chip that is used in a multi-function computer or device with the help of multiple chips to handle various tasks. On the other hand, a microcontroller is more self-contained as it has CPU, RAM, ROM, I/O ports and timers on the same chip; functions independently as a single-chip computer; and is dedicated to a specific task to control a single system. Further, it is often part of an embedded system.

Answers compiled by Col (Retd) N.C. Pande

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set of instructions to carry out user-defined tasks. It can also access external memory chips to both read and write data from and to the memory. The name itself conveys that it is small (micro) and will do any computation (processor).

From architectural point of view, a microprocessor features an arithmetic and logic unit, accumulator register,

is to control (controller) objects, processes or events. It is dedicated to one task or a set of closely related tasks.

A microcontroller is similar to a personal computer, as it has similar components built onto a single chip; it has its own CPU, memory (ROM and RAM), I/O ports, serial ports and timer.

Depending upon the architecture,

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